GATE ME 90 Days Challenge

Day 29/90 · Practice Set with Detailed Solutions

SUBJECT	TODAY'S TOPIC	COUNTDOWN
Fluid Mechanics	**Fluid Kinematics & Bernoulli**	**78 days to GATE**

How to use this set: 15 questions (6 MCQ, 9 NAT) on streamlines, stream function and velocity potential, continuity and control-volume mass balance, acceleration of fluid particles, vorticity and circulation, basic potential flows, and Bernoulli's equation. Try them in **45 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE.

Today's test: open 7 to 9 PM in the **GATE ME 90 Days Challenge** course on our app and website (enrol once for ₹1: iurlk.courses.store/909124), 20 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

Questions

Q1 MCQ

The streamlines of the two-dimensional flow $u = x$, $v = -y$ are

(A) straight lines through the origin
(B) rectangular hyperbolas xy = constant
(C) circles centred at the origin
(D) parabolas y = constant $\times x^2$

Q2 NAT

For a three-dimensional incompressible flow, $u = 3x^2$ and $v = -2xy$, and $w = 0$ on the plane $z = 0$. The value of w at the point (1, 1, 2) is _____

Q3 NAT

The stream function of a two-dimensional flow is $\psi = 3xy$ (SI units). The speed at the point (2, 1) (in m/s, round off to two decimal places) is _____ (Take $u = \partial\psi/\partial y$, $v = -\partial\psi/\partial x$ and $u = \partial\phi/\partial x$, $v = \partial\phi/\partial y$.)

Q4 MCQ

A velocity potential ϕ exists for a flow only if the flow is

(A) incompressible
(B) steady
(C) irrotational
(D) two-dimensional

Q5 NAT

The velocity potential of a two-dimensional flow is $\phi = 3xy$ (SI units). If the stream function is zero at the origin, its value at the point (1, 3) is _____ (Take $u = \partial\psi/\partial y$, $v = -\partial\psi/\partial x$ and $u = \partial\phi/\partial x$, $v = \partial\phi/\partial y$.)

Q6 NAT

A two-dimensional unsteady flow has $u = 2xt$ and $v = -2yt$ (SI units). The magnitude of the acceleration of a fluid particle at the point (1, 2) at $t = 1$ s (in m/s², round off to two decimal places) is _____

Q7 MCQ

Which statement about a flow net is **NOT** correct?

(A) Streamlines and equipotential lines intersect at right angles

(B) The discharge between any two adjacent streamlines is the same

(C) Where streamlines crowd together the velocity is high

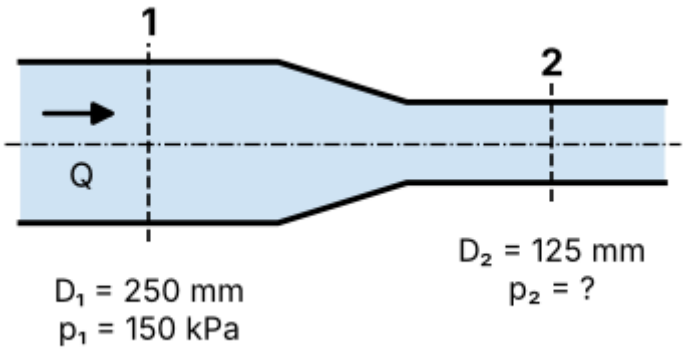
(D) A flow net can be drawn for rotational flows near solid boundaries

Q8 NAT

For the velocity field $u = -3y$, $v = 3x$ (SI units), the circulation around a circle of radius 0.5 m centred at the origin (in m^2/s, round off to two decimal places) is _____

Q9 NAT

Water flows at 0.08 m^3/s through the horizontal pipe shown, which tapers from 250 mm diameter at section 1 to 125 mm at section 2. The gauge pressure at section 1 is 150 kPa. Neglecting losses, the gauge pressure at section 2 (in kPa, round off to two decimal places) is _____



Q10 NAT

At a point in a pipeline carrying water, the velocity is 4 m/s, the gauge pressure is 40 kPa and the elevation above the datum is 6 m. The total energy head at the point (in m of water, round off to two decimal places) is _____

Q11 MCQ

A water jet issues from a nozzle at 20 m/s, inclined at 60° to the horizontal. Neglecting air resistance, the maximum height reached by the jet above the nozzle is

(A) 15.29 m

(B) 20.39 m

(C) 5.10 m

(D) 17.66 m

Q12 NAT

Air leaks from a rigid tank of volume 0.5 m^3 through a small hole of area 50 mm^2. At an instant the air in the hole has a density of 6 kg/m^3 and a velocity of 200 m/s. The rate of change of the density of the air in the tank at that instant (in kg/m^3 per s, round off to two decimal places) is _____

Q13 MCQ

The flow past a long circular cylinder (without circulation) can be obtained by superposing a uniform flow on a

(A) source

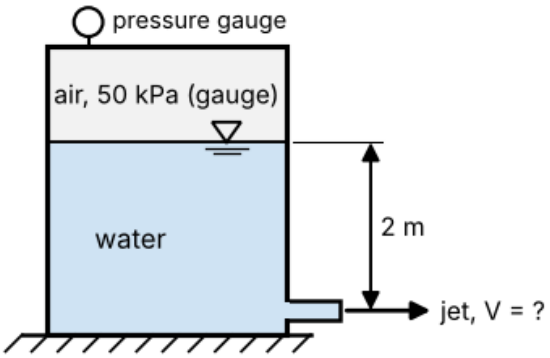
(B) doublet

(C) free vortex

(D) sink

Q14 NAT

A closed tank contains water with air at 50 kPa (gauge) above it. Water discharges to the atmosphere through a small nozzle 2 m below the water surface, as shown. Neglecting losses, the velocity of the jet (in m/s, round off to two decimal places) is _____



Q15 MCQ

Which statement about a free (irrotational) vortex is correct?

(A) The tangential velocity varies inversely with the radius, and Bernoulli's constant is the same for all streamlines

(B) The tangential velocity is proportional to the radius, and the free surface is a paraboloid

(C) The vorticity is uniform and equal to twice the angular velocity

(D) The pressure decreases with increasing radius

Answer Key

Q	Type	Answer
Q1	MCQ	(B) rectangular hyperbolas xy = constant
Q2	NAT	-8
Q3	NAT	6.71 (accept 6.67 to 6.75)
Q4	MCQ	(C) irrotational
Q5	NAT	12
Q6	NAT	7.21 (accept 7.17 to 7.25)
Q7	MCQ	(D) A flow net can be drawn for rotational flows near solid boundaries
Q8	NAT	4.71 (accept 4.68 to 4.74)
Q9	NAT	130.08 (accept 129.43 to 130.73)
Q10	NAT	10.89 (accept 10.84 to 10.95)
Q11	MCQ	(A) 15.29 m
Q12	NAT	-0.12 (accept −0.13 to −0.11)
Q13	MCQ	(B) doublet
Q14	NAT	11.80 (accept 11.74 to 11.86)
Q15	MCQ	(A) The tangential velocity varies inversely with the radius, and Bernoulli's constant is the same for all streamlines

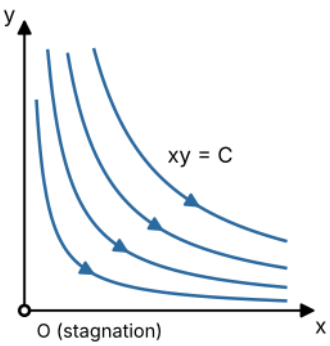
Detailed Solutions

Q1 **Answer: (B) rectangular hyperbolas xy = constant**

Concept: A streamline is everywhere tangent to the velocity: $dx/u = dy/v$.

$dx/x = dy/(-y) \rightarrow \ln x = -\ln y + c \rightarrow$ **xy = constant** (rectangular hyperbolas, with a stagnation point at the origin; see the sketch).

Exam tip: This is the flow towards a flat wall (stagnation-point flow). Its ψ is xy, and its φ is $(x^2 - y^2)/2$.



Q2 **Answer: -8**

Concept: Continuity for incompressible flow: $\partial u/\partial x + \partial v/\partial y + \partial w/\partial z = 0$.

$\partial u/\partial x + \partial v/\partial y = 6x - 2x = 4x$, so $\partial w/\partial z = -4x$ and $w = -4xz + f(x, y)$.

$w = 0$ at $z = 0$ gives $f = 0$, so $w = -4xz = -4 \times 1 \times 2 =$ **-8**

Exam tip: When integrating with respect to z, the 'constant' can be a function of x and y; use the boundary condition to fix it.

Q3 **Answer: 6.71 (accept 6.67 to 6.75)**

Concept: Velocity components from the stream function: $u = \partial\psi/\partial y$, $v = -\partial\psi/\partial x$.

$u = 3x = 6$ m/s, $v = -3y = -3$ m/s

$|V| = \sqrt{(6^2 + (-3)^2)} = \sqrt{45} =$ **6.71 m/s**

Exam tip: Check the sign convention in the question; some books use $u = -\partial\psi/\partial y$. The speed comes out the same either way.

Q4 **Answer: (C) irrotational**

Concept: If $V = \nabla\varphi$, then curl V = curl$(\nabla\varphi) = 0$ identically. So φ can exist only where the vorticity is zero.

Compressibility, steadiness or dimension do not matter; the flow must be **irrotational**.

Exam tip: ψ needs continuity (2-D); φ needs irrotationality. Both exist together only for 2-D, incompressible, irrotational (potential) flow, and then both satisfy Laplace's equation.

Q5 **Answer: 12**

Concept: Use $u = \partial\varphi/\partial x = \partial\psi/\partial y$ and $v = \partial\varphi/\partial y = -\partial\psi/\partial x$ (Cauchy–Riemann relations).

$u = 3y = \partial\psi/\partial y \rightarrow \psi = 1.5y^2 + f(x)$

$v = 3x = -\partial\psi/\partial x = -f'(x) \rightarrow f = -1.5x^2$; so $\psi = 1.5(y^2 - x^2)$

$\psi(1, 3) = 1.5(9 - 1) =$ **12**

Exam tip: Lines of constant φ (hyperbolas $xy = c$) and constant ψ (hyperbolas $y^2 - x^2 = c$) cross at right angles: a flow net.

Q6 **Answer: 7.21 (accept 7.17 to 7.25)**

Concept: Total acceleration = local $(\partial V/\partial t)$ + convective $(u\,\partial V/\partial x + v\,\partial V/\partial y)$.

$a_x = \partial u/\partial t + u\,\partial u/\partial x + v\,\partial u/\partial y = 2x + (2xt)(2t) + 0 = 2 + 4 =$ **6 m/s²**

$a_y = \partial v/\partial t + u\,\partial v/\partial x + v\,\partial v/\partial y = -2y + 0 + (-2yt)(-2t) = -4 + 8 =$ **4 m/s²**

$|a| = \sqrt{(6^2 + 4^2)} =$ **7.21 m/s²**

Exam tip: Don't drop the local term: in unsteady flow $\partial u/\partial t$ is non-zero even at a fixed point.

Q7 **Answer: (D) A flow net can be drawn for rotational flows near solid boundaries**

Concept: A flow net is the grid of streamlines (ψ = const) and equipotential lines (φ = const). It needs φ, so the flow must be irrotational (potential flow).

(A), (B) and (C) are true: the two families are orthogonal, the discharge between adjacent streamlines is constant, and closer streamlines mean higher velocity.

(D) is false: near walls viscous effects make the flow rotational (boundary layer), where no φ and hence no flow net exists. **Answer (D)**

Exam tip: Flow nets are used for seepage under dams and for flow outside the boundary layer.

Q8 **Answer: 4.71 (accept 4.68 to 4.74)**

Concept: Stokes' theorem: circulation $\Gamma = \oint V \cdot dr = \iint \omega_z \, dA$, with vorticity $\omega_z = \partial v/\partial x - \partial u/\partial y$.

$\omega_z = 3 - (-3) =$ **$6\ s^{-1}$** (uniform: a forced vortex rotating at 3 rad/s)

$\Gamma = 6 \times \pi(0.5)^2 =$ **$4.71\ m^2/s$**

Check directly: on r = 0.5 the speed is 3r = 1.5 m/s, so $\Gamma = 1.5 \times 2\pi(0.5) = 4.71\ m^2/s$.

Exam tip: Angular velocity = vorticity/2 = 3 rad/s.

Q9 **Answer: 130.08 (accept 129.43 to 130.73)**

Concept: Continuity gives the velocities; Bernoulli ($z_1 = z_2$) then gives the pressure drop.

$V_1 = 0.08/(\pi/4 \times 0.25^2) =$ **$1.630\ m/s$**, $V_2 = 4V_1 =$ **$6.519\ m/s$**

$p_2 = p_1 - \rho(V_2^2 - V_1^2)/2 = 150000 - 500(42.50 - 2.66) =$ **$130.08\ kPa$**

Exam tip: Halving the diameter makes V four times larger and the kinetic head sixteen times larger.

Q10 **Answer: 10.89 (accept 10.84 to 10.95)**

Concept: Total head H = p/(ρg) + V²/(2g) + z (pressure head + velocity head + datum head).

p/(ρg) = 40000/9810 = 4.077 m; V²/(2g) = 16/19.62 = 0.815 m; z = 6 m

H = **10.89 m**

Exam tip: The hydraulic grade line is p/ρg + z (here 10.08 m); the energy line is a velocity head above it.

Q11 **Answer: (A) 15.29 m**

Concept: In a free jet the pressure is atmospheric throughout and the horizontal velocity component stays constant. At the top only V cos θ remains.

Bernoulli between the nozzle and the top: $V^2/2g = (V\cos\theta)^2/2g + h \rightarrow h = V^2\sin^2\theta/(2g)$

h = 400 × 0.75/19.62 = **15.29 m**

Exam tip: 20.39 m is the height for a vertical jet (θ = 90°); 5.10 m uses cos²θ by mistake.

Q12 **Answer: −0.12 (accept −0.13 to −0.11)**

Concept: Unsteady control-volume mass balance: $d(\rho V_{tank})/dt + \dot{m}_{out} = 0$.

$\dot{m}_{out} = \rho AV = 6 \times 50 \times 10^{-6} \times 200 =$ **$0.06\ kg/s$**

$d\rho/dt = -\dot{m}_{out}/V_{tank} = -0.06/0.5 =$ **$-0.12\ kg/m^3$ per s** (density falling)

Exam tip: The sign matters in NAT: a decreasing density gives a negative answer.

Q13 **Answer: (B) doublet**

Concept: A doublet is a source and sink of equal strength brought together. Added to a uniform flow U, the streamline ψ = 0 becomes a circle of radius $a = \sqrt{(\mu/2\pi U)}$.

Uniform flow + source → Rankine half-body; uniform flow + source + sink (separated) → Rankine oval; uniform flow + **doublet** → circular cylinder; adding a free vortex gives lift (Magnus effect).

Exam tip: On the cylinder surface the speed is 2U sin θ: maximum 2U at the top and bottom, zero at the two stagnation points.

Q14 **Answer: 11.80 (accept 11.74 to 11.86)**

Concept: Bernoulli from the water surface (V ≈ 0, p = 50 kPa gauge) to the jet (p = 0 gauge): V = √[2(p/ρ + gh)].

The air pressure is equivalent to an extra head of 50000/9810 = **5.10 m** of water, so the effective head is 7.10 m.

V = √[2(50 + 19.62)] = √139.24 = **11.80 m/s**

Exam tip: Ignoring the air pressure gives √(2g × 2) = 6.26 m/s, a common slip.

Q15 **Answer: (A) The tangential velocity varies inversely with the radius, and Bernoulli's constant is the same for all streamlines**

Concept: Free vortex: no external torque, so angular momentum is conserved: V r = constant (V ∝ 1/r). The flow is irrotational everywhere except at the centre.

Because it is irrotational, $p/\rho + V^2/2 + gz$ is the **same constant on every streamline**. In a forced vortex (V = ωr) the flow is rotational and the constant changes from one streamline to another.

Exam tip: Examples: free vortex — water draining from a basin, flow in a whirlpool chamber; forced vortex — a liquid rotating with its container (paraboloid surface).

What next?

Day 29 test tonight, 7 to 9 PM

20 questions · 60 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with name, marks and photo (if shared) in the next 8 AM challenge reel. Not enrolled yet? Join the ₹1 challenge course: iurlk.courses.store/909124

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